

BJT DC BIASING – COMPLETE FORMULA SHEET

0. UNIVERSAL RELATIONS (Always Applicable)

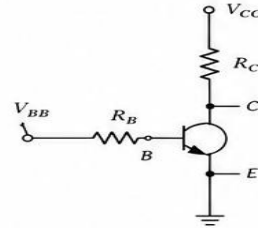
$$I_C = \beta I_B$$

$$I_E = (\beta + 1) I_B$$

$$V_{BE} \approx 0.7 \text{ V (Si)}$$

$$V_{CE} = V_C - V_E$$

1. FIXED BIAS (BASE BIAS)



Base loop :

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

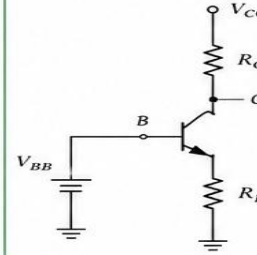
Collector :

$$I_C = \beta I_B$$

$$V_C = V_{CC} - I_C R_C$$

$$V_{CE} = V_C \quad (\text{since } V_E = 0)$$

2. EMITTER BIAS



Emitter loop :

$$I_E = \frac{V_{BB} - V_{BE}}{R_E}$$

Approximation :

$$I_C \approx I_E$$

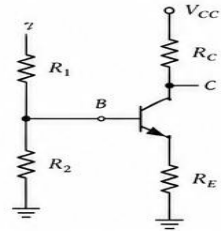
Voltages :

$$V_E = I_E R_E$$

$$V_C = V_{CC} - I_C R_C$$

$$V_{CE} = V_C - V_E$$

3. VOLTAGE DIVIDER BIAS (MOST IMPORTANT)



Step 1: Thevenin Equivalent

$$R_{th} = R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

$$V_{th} = \frac{R_2}{R_1 + R_2} V_{CC}$$

Step 2: Base current

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E}$$

Step 3: Currents

$$I_C = \beta I_B$$

$$I_E = (\beta + 1) I_B$$

Step 4: Voltages

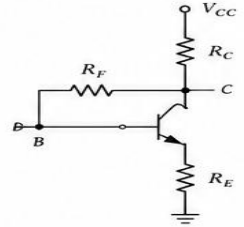
$$V_E = I_E R_E$$

$$V_B = V_E + V_{BE}$$

$$V_C = V_{CC} - I_C R_C$$

$$V_{CE} = V_C - V_E$$

4. COLLECTOR-FEEDBACK BIAS



Base current :

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta (R_C + R_E)}$$

Currents :

$$I_C = \beta I_B$$

$$I_E = (\beta + 1) I_B$$

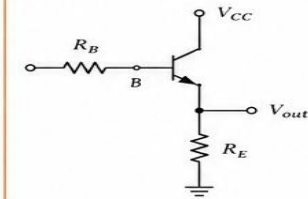
Voltages :

$$V_E = I_E R_E$$

$$V_C = V_{CC} - I_C R_C$$

$$V_{CE} = V_C - V_E$$

5. EMITTER FOLLOWER (COMMON COLLECTOR)



Base loop :

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E}$$

Currents :

$$I_E = (\beta + 1) I_B$$

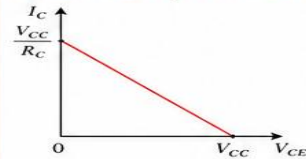
$$I_C \approx I_E$$

Voltages :

$$V_E = I_E R_E$$

$$V_{out} = V_E$$

6. LOAD LINE (VERY IMPORTANT CHECK)



$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$

Key points:

- Cutoff: $I_C = 0$, $V_{CE} = V_{CC}$
- Maximum current:

$$I_{C(max)} = \frac{V_{CC}}{R_C}$$

7. REALITY CHECK (No saturation voltage needed)

Compare :

$$I_C \text{ (calculated)} \text{ vs } \frac{V_{CC}}{R_C}$$

Decision:

- If $I_C < \frac{V_{CC}}{R_C} \rightarrow$ valid (active region)
- If $I_C > \frac{V_{CC}}{R_C} \rightarrow$ wrong assumption (circuit limit reached)

8. QUICK REFERENCE TABLE

Circuit Type	Key Idea	Key Formula (Start Here)	Stability
Fixed Bias (Base Bias)	Base from fixed supply	$I_B = (V_{BB} - 0.7)/R_B$	Poor
Emitter Bias	Emitter resistor stabilizes current	$I_E = (V_{BB} - 0.7)/R_E$	Better
Voltage Divider Bias	Base from divider (most used)	$I_B = \frac{V_{th} - 0.7}{R_{th} + (\beta + 1)R_E}$	Good
Collector-Feedback Bias	Negative feedback via R_F	$I_B = \frac{V_{CC} - 0.7}{R_F + \beta(R_C + R_E)}$	Good
Emitter Follower	Output at emitter (buffer)	$I_B = \frac{V_{CC} - 0.7}{R_B + (\beta + 1)R_E}$	Very Good

9. REMEMBER (Always!)

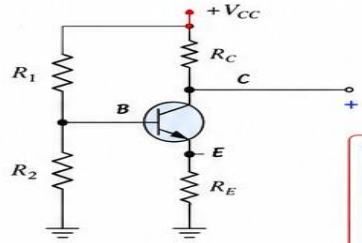
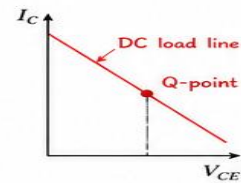
- ✓ DC analysis $\rightarrow$ Capacitors OPEN
- ✓ AC midband $\rightarrow$ Capacitors SHORT
- ✓ Solve bias first, then check limits
- ✓ Always check load line
- ✓ Physical understanding > memorization

BJT AMPLIFIER – CONCEPTUAL FORMULA SHEET (DC + AC)

0. BIG PICTURE (ALWAYS START HERE)

Two separate analyses:

- **DC** → Bias (Q-point)
- **AC** → Gain + Impedance



1. DC ANALYSIS (BIASING)

Capacitors are OPEN circuit

Steps:

1. Remove all capacitors
2. Solve using DC circuits only

Voltage Divider → Thevenin:

$$V_{th} = \frac{R_2}{R_1 + R_2} V_{CC}$$

$$R_{th} = R_1 \parallel R_2$$

Base loop:

$$I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1)R_E}$$

Core relations:

$$I_C = \beta I_B$$

$$I_E = (\beta + 1) I_B$$

$$V_{BE} \approx 0.7 \text{ V}$$

Voltages:

$$V_E = I_E R_E$$

$$V_B = V_E + V_{BE}$$

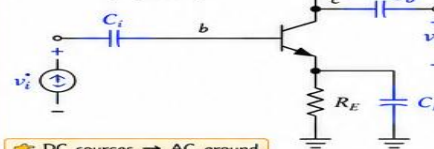
$$V_C = V_{CC} - I_C R_C$$

$$V_{CE} = V_C - V_E$$

2. AC ANALYSIS (MIDBAND MODEL)

Capacitors are SHORT circuit

- C_i → input short
- C_o → output short
- C_E → emitter bypass (if present)



DC sources → AC ground

3. SMALL-SIGNAL MODEL

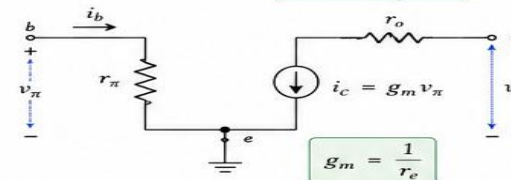
Thermal voltage:

$$V_T \approx 26 \text{ mV}$$

Key parameters:

$$r_e = \frac{26 \text{ mV}}{I_E}$$

$$r_\pi = \beta r_e$$



4. VOLTAGE GAIN A_v

Case 1: Emitter bypass capacitor present (C_E shorted)

$$A_v = - \frac{R_C \parallel r_o}{r_e}$$

If $r_o \rightarrow \infty$:

$$A_v \approx - \frac{R_C}{r_e}$$

Case 2: NO bypass capacitor

$$A_v = - \frac{R_C}{r_e + R_E}$$

(Negative sign → 180° phase shift)

5. INPUT IMPEDANCE Z_i

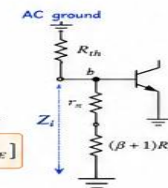
Looking through C_i

General:

$$Z_i = R_{th} \parallel r_\pi$$

If emitter resistor NOT bypassed:

$$Z_i = R_{th} \parallel [r_\pi + (\beta + 1)R_E]$$



7. OUTPUT VOLTAGE

$$v_o = A_v \cdot v_i$$

→ Phase:

Common-emitter
→ 180° phase shift

8. CAPACITOR ROLES (CONCEPTUAL)

Capacitor	Role
C_i	Blocks DC, passes AC input
C_o	Blocks DC, passes output
C_E	Increases gain (removes R_E in AC)

9. KEY DESIGN INSIGHT (VERY IMPORTANT)

Gain depends on:

- Small r_e → high gain
- Large R_C → high gain

Input impedance depends on:

- r_π
- Bias network loading

Output impedance:

- Mostly R_C

Recall:

$$r_e = \frac{26 \text{ mV}}{I_E}$$

↓ I_E (too small)
→ r_e large
→ gain low

↑ I_E (too large)
→ distortion
& power waste

10. COMMON STUDENT MISTAKES (TEACH THIS!)

- ✗ Using DC formulas in AC analysis
- ✗ Forgetting capacitors:
 - DC → open
 - AC → short
- ✗ Mixing r_e and R_E
- ✗ Forgetting $Z_i = R_{th} \parallel r_\pi$



ONE-LINE SUMMARY (GIVE STUDENTS THIS)

DC sets the operating point. **AC** determines how the signal behaves around that point.

QUESTION 1 (20 MARKS)

The NPN transistor circuit shown in Figure 1 uses a fixed-bias (base-bias) configuration.

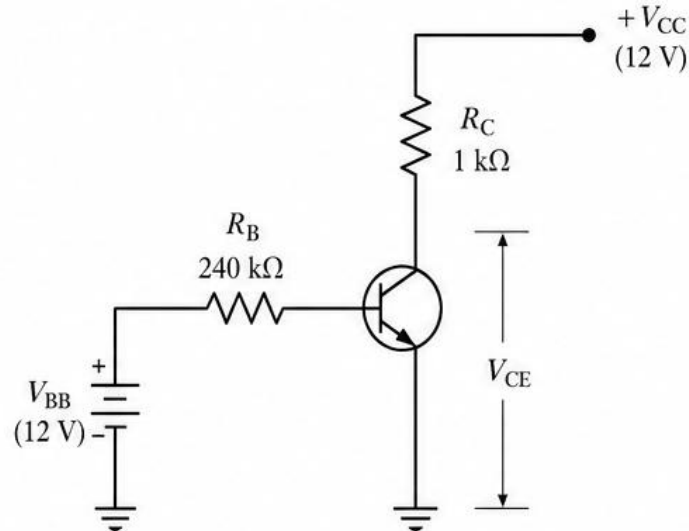


Figure 1

- Calculate the base current I_B . (3 marks)
- If $\beta = 100$, calculate the collector current I_C . (3 marks)
- Calculate the collector-emitter voltage V_{CE} . (3 marks)
- Determine the region of operation. (2 marks)
- Determine the saturation current $I_{C(sat)}$ and the cutoff voltage $V_{CE(cutoff)}$. (4 marks)
- Draw the DC load line on the output characteristics axes shown in Figure 2 and indicate the Q-point corresponding to the values found in (b) and (c). (5 marks)

Assume $V_{BE} = 0.7$ V.

SOLUTION

Step 1: Calculate I_B

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{12 - 0.7}{240\,000}$$

$$= \frac{11.3}{240\,000} = 47.1\, \mu\text{A}$$

Step 2: Calculate I_C (active region assumption)

$$I_C = \beta I_B = 100 \times 47.1\, \mu\text{A} = 4.71\, \text{mA}$$

Step 3: Calculate V_{CE}

$$V_{CE} = V_{CC} - I_C R_C = 12 - (4.71\, \text{mA})(1\, \text{k}\Omega)$$

$$= 12 - 4.71 = 7.29\, \text{V}$$

Step 4: Region of operation

Since $V_{CE} = 7.29\, \text{V} > 0.2\, \text{V}$ (approximately), transistor is in the **ACTIVE** (forward-active) region.

Step 5: Determine $I_{C(sat)}$ and $V_{CE(cutoff)}$

- Saturation current ($I_{C(sat)}$):**

When $V_{CE} = 0$ (mental short),

$$I_{C(sat)} = \frac{V_{CC}}{R_C} = \frac{12}{1\,000} = 12\, \text{mA}$$

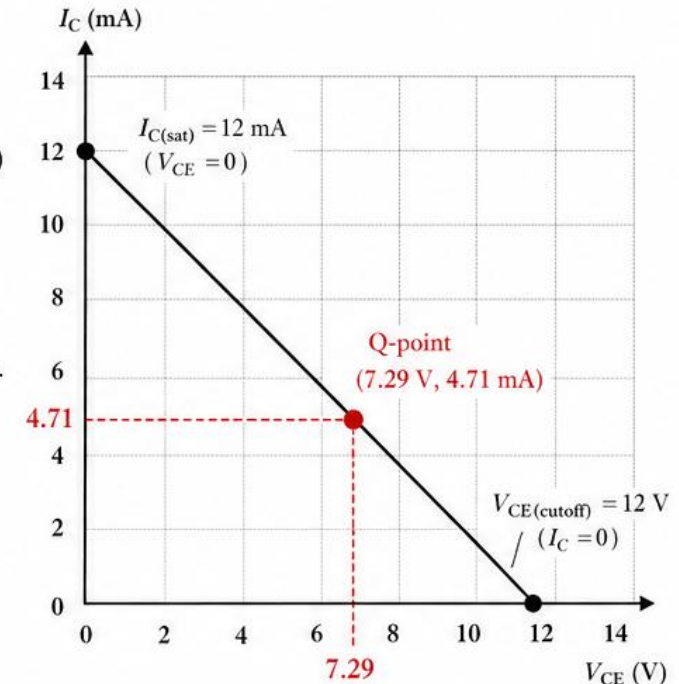
- Cutoff voltage ($V_{CE(cutoff)}$):**

When $I_C = 0$ (mental open), $V_{CE(cutoff)} = V_{CC} = 12\, \text{V}$.

Step 6: DC load line and Q-point (Figure 2)

- Plot points (0, 12 mA) and (12 V, 0).
- Draw straight line between them (load line).
- Q-point: ($V_{CE} = 7.29\, \text{V}$, $I_C = 4.71\, \text{mA}$).

Figure 2: Output Characteristics with DC Load Line



Final Answers

(a) $I_B = 47.1\, \mu\text{A}$	(d) Region: ACTIVE
(b) $I_C = 4.71\, \text{mA}$	(e) $I_{C(sat)} = 12\, \text{mA}$, $V_{CE(cutoff)} = 12\, \text{V}$
(c) $V_{CE} = 7.29\, \text{V}$	(f) Load line and Q-point shown in Figure 2

QUESTION (20 MARKS)

For the emitter-bias network shown in Figure 1, determine the following:

- a. I_B
- b. I_C
- c. V_{CE}
- d. V_C
- e. V_E
- f. V_B
- g. V_{BC}

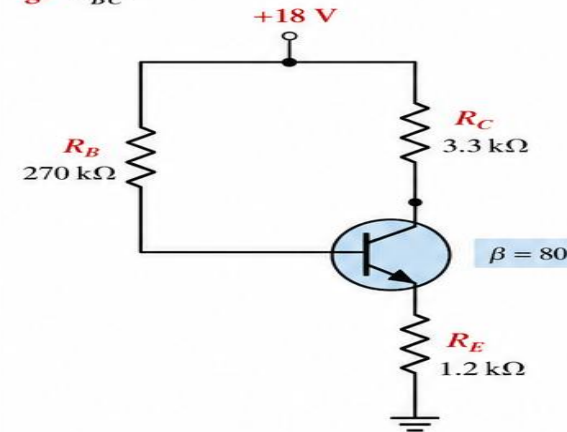


Figure 1

Given:

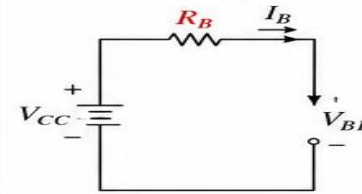
- $V_{CC} = 18 \text{ V}$
- $R_B = 270 \text{ k}\Omega$
- $R_C = 3.3 \text{ k}\Omega$
- $R_E = 1.2 \text{ k}\Omega$
- $\beta = 80$
- Assume $V_{BE} = 0.7 \text{ V}$

Instruction:

Show all steps clearly.
Round final answers appropriately.

SOLUTION

Step 1: Find I_B using KVL in base-emitter loop



$$V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

$$I_E = (\beta + 1) I_B$$

$$\begin{aligned} \Rightarrow I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} \\ &= \frac{18 - 0.7}{270 \text{ k}\Omega + (80 + 1)(1.2 \text{ k}\Omega)} \\ &= \frac{17.3}{367.2 \text{ k}\Omega} \end{aligned}$$

$$I_B = 47.12 \mu\text{A}$$

Step 2: Find I_C and I_E

$$I_C = \beta I_B = 80 \times 47.12 \mu\text{A}$$

$$I_C = 3.77 \text{ mA}$$

$$I_E = (\beta + 1) I_B = 81 \times 47.12 \mu\text{A}$$

$$I_E = 3.82 \text{ mA}$$

Step 3: Find V_E

$$V_E = I_E R_E = (3.82 \text{ mA})(1.2 \text{ k}\Omega)$$

$$V_E = 4.58 \text{ V}$$

Step 4: Find V_B

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 4.58 \text{ V}$$

$$V_B = 5.28 \text{ V}$$

Step 5: Find V_C using KVL in collector-emitter loop

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

$$\begin{aligned} V_C &= V_{CC} - I_C R_C \\ &= 18 - (3.77 \text{ mA})(3.3 \text{ k}\Omega) \end{aligned}$$

$$V_C = 5.56 \text{ V}$$

Step 6: Find V_{CE}

$$V_{CE} = V_C - V_E = 5.56 \text{ V} - 4.58 \text{ V}$$

$$V_{CE} = 0.98 \text{ V}$$

Step 7: Find V_{BC}

$$V_{BC} = V_B - V_C = 5.28 \text{ V} - 5.56 \text{ V}$$

$$V_{BC} = -0.28 \text{ V}$$

FINAL ANSWERS

$$\text{a. } I_B = 47.12 \mu\text{A}$$

$$\text{b. } I_C = 3.77 \text{ mA}$$

$$\text{c. } V_{CE} = 0.98 \text{ V}$$

$$\text{d. } V_C = 5.56 \text{ V}$$

$$\text{e. } V_E = 4.58 \text{ V}$$

$$\text{f. } V_B = 5.28 \text{ V}$$

$$\text{g. } V_{BC} = -0.28 \text{ V}$$

Note: $V_{CE} > 0$ (very small), so the transistor is in the **ACTIVE** region (near saturation but still active).

QUESTION (20 MARKS)

For the voltage-divider bias circuit shown in Figure 1, determine:

- The Thevenin resistance R_{Th} seen by the base.
- The Thevenin voltage E_{Th} .
- The base current I_B .
- The collector current I_C .
- The dc collector-emitter voltage V_{CE} .

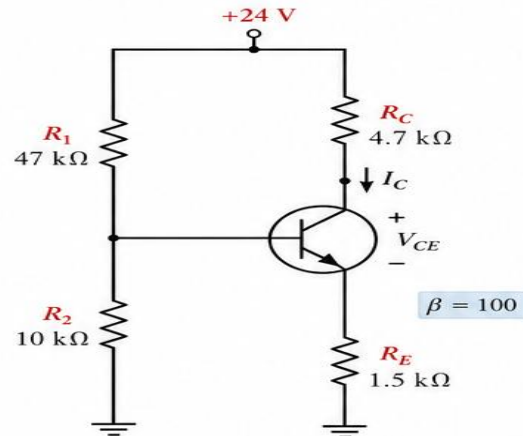


Figure 1

Given:

- $V_{CC} = 24 \text{ V}$
- $R_1 = 47 \text{ k}\Omega$
- $R_2 = 10 \text{ k}\Omega$
- $R_C = 4.7 \text{ k}\Omega$
- $R_E = 1.5 \text{ k}\Omega$
- $\beta = 100$
- Assume $V_{BE} = 0.7 \text{ V}$

Instruction:

Show all steps clearly.
Round final answers appropriately.

SOLUTION**Step 1: Find R_{Th} (Thevenin resistance seen by the base)**

$$\begin{aligned} R_{Th} &= R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2} \\ &= \frac{(47 \text{ k}\Omega)(10 \text{ k}\Omega)}{47 \text{ k}\Omega + 10 \text{ k}\Omega} \\ &= \frac{470}{57} \text{ k}\Omega = 8.25 \text{ k}\Omega \end{aligned}$$

$$R_{Th} = 8.25 \text{ k}\Omega$$

Step 2: Find E_{Th} (Thevenin voltage at the base node)

$$\begin{aligned} E_{Th} &= \frac{R_2}{R_1 + R_2} V_{CC} \\ &= \frac{10 \text{ k}\Omega}{47 \text{ k}\Omega + 10 \text{ k}\Omega} (24 \text{ V}) \\ &= \frac{10}{57} (24 \text{ V}) = 4.21 \text{ V} \end{aligned}$$

$$E_{Th} = 4.21 \text{ V}$$

Step 3: Find I_B (KVL in base-emitter loop)

$$\begin{aligned} I_B &= \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \\ &= \frac{4.21 \text{ V} - 0.7 \text{ V}}{8.25 \text{ k}\Omega + (100 + 1)(1.5 \text{ k}\Omega)} \\ &= \frac{3.51 \text{ V}}{8.25 \text{ k}\Omega + 151.5 \text{ k}\Omega} \\ &= \frac{3.51}{159.75 \text{ k}\Omega} = 21.97 \mu\text{A} \end{aligned}$$

$$I_B = 21.97 \mu\text{A}$$

Step 4: Find I_C

$$I_C = \beta I_B = 100(21.97 \mu\text{A}) = 2.20 \text{ mA}$$

$$I_C = 2.20 \text{ mA}$$

Step 5: Find V_{CE} (KVL in collector-emitter loop)

$$\begin{aligned} V_{CE} &= V_{CC} - I_C(R_C + R_E) \\ &= 24 \text{ V} - (2.20 \text{ mA})(4.7 \text{ k}\Omega + 1.5 \text{ k}\Omega) \\ &= 24 \text{ V} - (2.20 \text{ mA})(6.2 \text{ k}\Omega) \\ &= 24 \text{ V} - 13.64 \text{ V} \\ &= 10.36 \text{ V} \end{aligned}$$

$$V_{CE} = 10.36 \text{ V}$$

FINAL ANSWERS

$$\text{a. } R_{Th} = 8.25 \text{ k}\Omega$$

$$\text{b. } E_{Th} = 4.21 \text{ V}$$

$$\text{c. } I_B = 21.97 \mu\text{A}$$

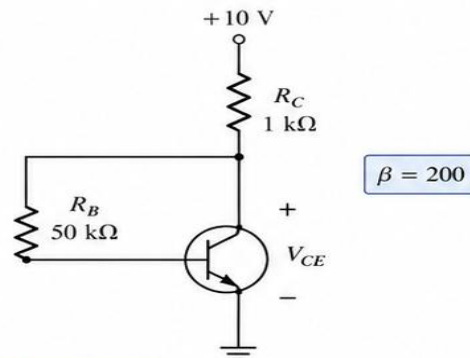
$$\text{d. } I_C = 2.20 \text{ mA}$$

$$\text{e. } V_{CE} = 10.36 \text{ V}$$

Note: Since $V_{CE} = 10.36 \text{ V} > 0.2 \text{ V}$, the transistor is in the **ACTIVE** region.

PROBLEM:

A fixed-bias BJT circuit is given below.

**GIVEN:**

- $V_{CC} = 10\text{ V}$
- $R_C = 1\text{ k}\Omega$
- $R_B = 50\text{ k}\Omega$
- $\beta = 200$
- $V_{BE} = 0.7\text{ V}$

IMPORTANT:

Do NOT use $V_{CE(sat)}$.
Use only circuit limits
and load-line reasoning.

(a) Calculate base current I_B

Base is fed from V_{CC} through R_B . Assuming silicon transistor,
 $V_{BE} = 0.7\text{ V}$.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{10 - 0.7}{50\text{ k}\Omega} = \frac{9.3}{50\text{ k}\Omega} = 0.186\text{ mA} = 186\text{ }\mu\text{A}$$

$$I_B = 186\text{ }\mu\text{A}$$

(b) Assume active region and calculate I_C and V_{CE}

Assuming active region,

$$I_C = \beta I_B = 200(186\text{ }\mu\text{A}) = 37.2\text{ mA}$$

$$I_C = 37.2\text{ mA}$$

Now,

$$\begin{aligned} V_{CE} &= V_{CC} - I_C R_C \\ &= 10 - (37.2\text{ mA})(1\text{ k}\Omega) \\ &= 10 - 37.2 \\ &= -27.2\text{ V} \end{aligned}$$

$$V_{CE} = -27.2\text{ V (IMPOSSIBLE!)}$$

(c) Reality check (NO 0.2 V allowed): Is the transistor really in active region?

Find the maximum possible collector current using circuit limits

Maximum possible collector current

$$I_{C(\max)} = \frac{V_{CC}}{R_C} = \frac{10\text{ V}}{1\text{ k}\Omega} = 10\text{ mA}$$

Compare this with the calculated I_C from part (b):

- Calculated $I_C = 37.2\text{ mA}$
- Maximum possible $I_C = 10\text{ mA}$

Since $I_C(\text{calc}) > I_{C(\max)}$,
the transistor CANNOT deliver
37.2 mA. The assumption of
active region is INVALID.

Therefore, the transistor is FORCED to deliver the maximum current allowed by the circuit.

(d) If assumption is invalid, find the actual collector current I_C

The collector current cannot exceed the limit set by R_C and V_{CC} .

Thus, the actual collector current is

$$I_C = I_{C(\max)} = \frac{V_{CC}}{R_C} = 10\text{ mA}$$

(e) Find the actual V_{CE} using physical reasoning

If $I_C = 10\text{ mA}$ flows through $R_C = 1\text{ k}\Omega$,
the voltage drop across R_C is

$$V_{RC} = I_C R_C = (10\text{ mA})(1\text{ k}\Omega) = 10\text{ V}$$

$$V_{CE} = V_{CC} - V_{RC} = 10\text{ V} - 10\text{ V} = 0\text{ V}$$

(Collector is at 0 V, same as emitter which is grounded.)

This is the physical limit of the circuit.

FINAL ANSWERS**(a)**

$$I_B = 186\text{ }\mu\text{A}$$

(b)

$$\begin{aligned} I_C &= 37.2\text{ mA (assumed)} \\ V_{CE} &= -27.2\text{ V (impossible)} \end{aligned}$$

(c)

Not in active region.
Active assumption
is INVALID.

(d)

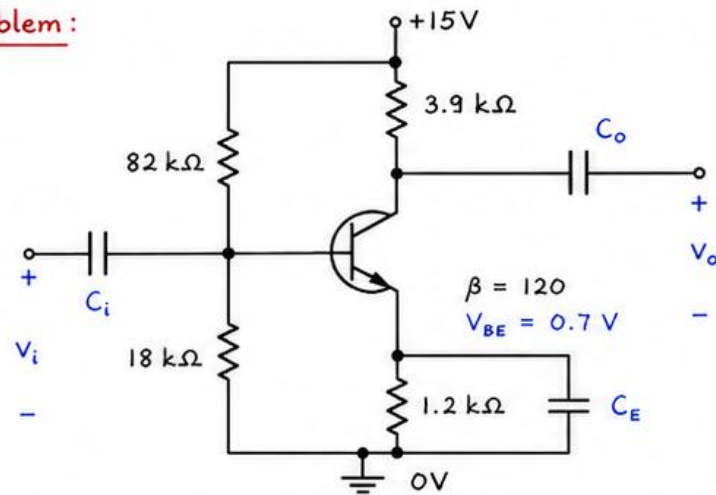
$$\text{Actual } I_C = 10\text{ mA}$$

(e)

$$\text{Actual } V_{CE} = 0\text{ V}$$

KEY TAKEAWAY: A transistor cannot deliver more current than the circuit can supply.
Always check against the maximum possible current V_{CC}/R_C (load-line limit).

Problem:



Given: $\beta = 120$, $V_{BE} = 0.7\text{ V}$

For the voltage-divider biased amplifier shown,

- 1) Find DC bias (I_B , I_C , V_E , V_C , V_{CE})
- 2) Find small-signal voltage gain A_v (midband)
- 3) Find input impedance Z_i (looking into base through C_i)
- 4) Find output impedance Z_o (looking into collector through C_o)
- 5) Find the output voltage v_o if $v_i = 10\text{ mV (rms)}$

Assume $r_o \rightarrow \infty$ and coupling/bypass capacitors are short at midband.

1) DC bias (I_B , I_C , V_E , V_C , V_{CE})

Thevenin seen at base from divider:

$$V_{TH} = 15 \times \frac{18}{82+18} = 15 \times \frac{18}{100} = 2.70\text{ V}$$

$$R_{TH} = 82\text{ k}\Omega \parallel 18\text{ k}\Omega = \frac{82 \times 18}{82+18} = 14.76\text{ k}\Omega$$

Base current:

$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta+1)R_E} = \frac{2.70 - 0.7}{14.76\text{ k} + (120+1)(1.2\text{ k})}$$

$$= \frac{2.00}{14.76\text{ k} + 145.2\text{ k}}$$

$$= \frac{2.00}{159.96\text{ k}} = 12.50\text{ }\mu\text{A}$$

$$I_B = 12.50\text{ }\mu\text{A}$$

$$I_C = \beta I_B = 120 (12.50\text{ }\mu\text{A}) = 1.50\text{ mA}$$

$$I_C = 1.50\text{ mA}$$

$$I_E = (\beta+1)I_B = 121 (12.50\text{ }\mu\text{A}) = 1.5125\text{ mA}$$

$$I_E = 1.513\text{ mA}$$

$$V_E = I_E R_E = (1.513\text{ mA})(1.2\text{ k}\Omega) = 1.816\text{ V}$$

$$V_E = 1.82\text{ V}$$

$$V_C = 15 - I_C R_C = 15 - (1.50\text{ mA})(3.9\text{ k}\Omega) = 9.15\text{ V}$$

$$V_C = 9.15\text{ V}$$

$$V_{CE} = V_C - V_E = 9.15 - 1.82 = 7.33\text{ V}$$

$$V_{CE} = 7.33\text{ V}$$

2) Small-signal voltage gain A_v (midband)

$$r_e = \frac{26\text{ mV}}{I_E} = \frac{26\text{ mV}}{1.513\text{ mA}} = 17.18\text{ }\Omega$$

$$R_C \parallel r_o = 3.9\text{ k}\Omega \quad (\text{since } r_o \rightarrow \infty)$$

$$A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{3.9\text{ k}\Omega}{17.18\text{ }\Omega}$$

$$= -227.0$$

$$A_v \approx -227$$

3) Input impedance Z_i (looking into base through C_i)

$$Z_i = R_{TH} \parallel r_{\pi}$$

$$\text{where } r_{\pi} = \beta r_e = 120 (17.18) = 2.062\text{ k}\Omega$$

$$Z_i = 14.76\text{ k}\Omega \parallel 2.062\text{ k}\Omega$$

$$= \frac{14.76\text{ k} \times 2.062\text{ k}}{14.76\text{ k} + 2.062\text{ k}}$$

$$= \frac{30.43 \times 10^6}{16.822\text{ k}}$$

$$= 1.809\text{ k}\Omega$$

$$Z_i \approx 1.81\text{ k}\Omega$$

4) Output impedance Z_o (looking into collector through C_o)

With input source set to zero $\rightarrow$ base is AC-grounded and $r_o \rightarrow \infty$, seen from collector is simply R_C .

$$Z_o = R_C \parallel r_o = 3.9\text{ k}\Omega \parallel \infty = 3.9\text{ k}\Omega$$

$$Z_o \approx 3.9\text{ k}\Omega$$

5) Output voltage v_o for $v_i = 10\text{ mV (rms)}$

$$v_o = A_v v_i = (-227)(10\text{ mV})$$

$$= -2270\text{ mV}$$

$$v_o \approx -2.27\text{ V (rms)}$$

(Output is 180° out of phase with input)

Final Answers:

$$I_B = 12.50\text{ }\mu\text{A}$$

$$I_C = 1.50\text{ mA}$$

$$V_E = 1.82\text{ V}$$

$$V_C = 9.15\text{ V}$$

$$V_{CE} = 7.33\text{ V}$$

$$A_v \approx -227$$

$$Z_i \approx 1.81\text{ k}\Omega$$

$$Z_o = 3.9\text{ k}\Omega$$

$$v_o \approx -2.27\text{ V (rms)}$$